Zero-Knowledge Proof based Blockchain Transaction Visualization with GNN Risk Prediction

Team members：

Shuwen Liu Zongni Jiang
Xinyu Quan Shanyou Dong

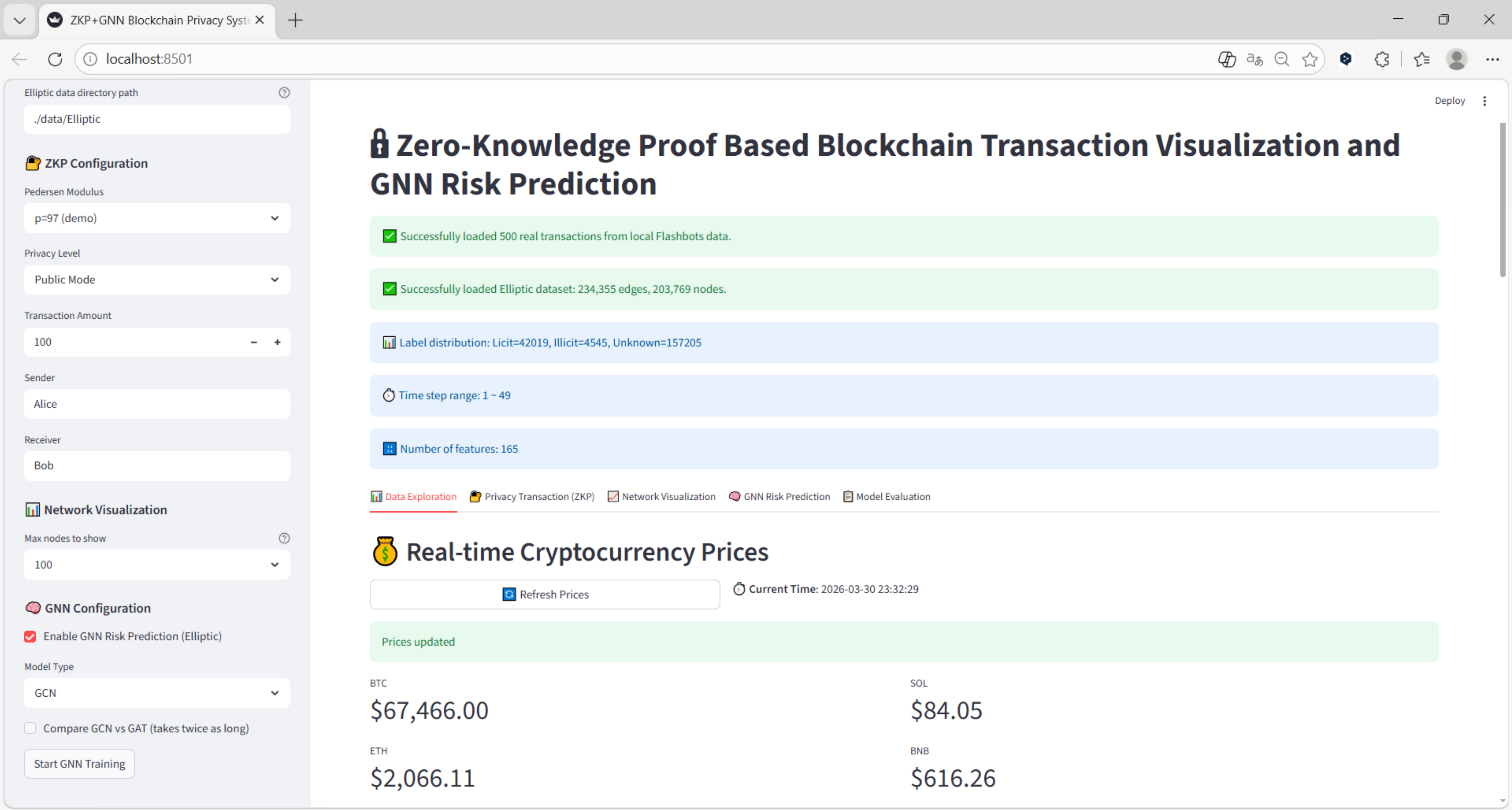
Abstract

厦門大學
XIAMEN UNIVERSITY

- We have developed an **interactive web application** that demonstrates privacy-preserving blockchain transactions implemented using zero-knowledge proofs, and performs P2P visual network analysis on transaction data. In addition, we have carried out GNN-based risk prediction using a **real-world transaction dataset**.
- The system has five tabs: **Data Exploration** (crypto prices and dataset stats), **Privacy Transaction** (ZKP demo with homomorphic verification), **Network Visualization** (P2P graphs with risk coloring), **GNN Risk Prediction** (train GCN/GAT on Elliptic dataset), and **Model Evaluation** (performance comparison).
- Users can adjust the **Pedersen modulus** (p=97 or industrial), privacy level (public or **ZKP** mode), transaction amount, sender/receiver names, GNN model type (**GCN** or **GAT**).

Abstract

Click to play the system tutorial video



Contents

厦門大學
XIAMEN UNIVERSITY

01

Project Background & Significance

Project Background

- **Technical characteristics:**
  The double-edged sword of trust and transparency

  The decentralised and immutable nature of blockchain technology has established a new mechanism of trust; however, its publicly accessible transaction ledger leaves users’ privacy completely exposed, and transaction histories can be traced at will.

- **Security Dilemma:**
  Privacy Breaches and the Risk of Fraud

  A fully transparent environment not only infringes on user privacy but is also exploited by criminals for activities such as money laundering and fraud; traditional rule-based detection struggles to cope with complex patterns of fund flows.

- **Key challenges:**
  privacy protection and risk identification

  How to effectively identify and block risky transactions on the blockchain whilst safeguarding user privacy is a key issue that urgently needs to be addressed in the field of blockchain security.

Schematic illustration of data flow and connectivity within a blockchain network

Project Significance and Objectives

XIAMEN UNIVERSITY

Significance

- **Theoretical significance**: Exploring the integration of zero-knowledge proofs with graph neural networks to provide new technical approaches for privacy protection and risk identification.
- **Practical value:** Providing tools for demonstrating private transactions and graph-based risk prediction models to enhance security analysis.
- **Educational value:** Visual interaction lowers the barrier to entry, presenting abstract cryptographic and machine learning concepts in an intuitive manner.

Objectives

1. Implement a non-interactive zero-knowledge proof system based on the Σ protocol, supporting the verification of private transactions.
2. Develop a transaction risk prediction model based on graph neural networks to classify nodes as legitimate or illegitimate.
3. Develop a Streamlit visual interface that integrates data exploration, training and evaluation capabilities.
4. To validate the system’s effectiveness, we compare the performance of the GCN and GAT models on real-world datasets.

厦門大學
XIAMEN UNIVERSITY

02

Data Description & Exploration

Data sources and pre-processing

Data sources

Module	Data source	Features	Loading method
ZKP-enabled private transactions	Flashbots Mempool Dumpster	Real transaction data, including key details such as addresses, amounts and timestamps	Local CSV/Parquet files; if none are found, use simulated data
GNN Risk Prediction	Elliptic Bitcoin Transaction Dataset	Labelled transaction graph data, comprising 166-dimensional node features and labels	Local read, with configurable path

Data pre-processing

Flashbots Data

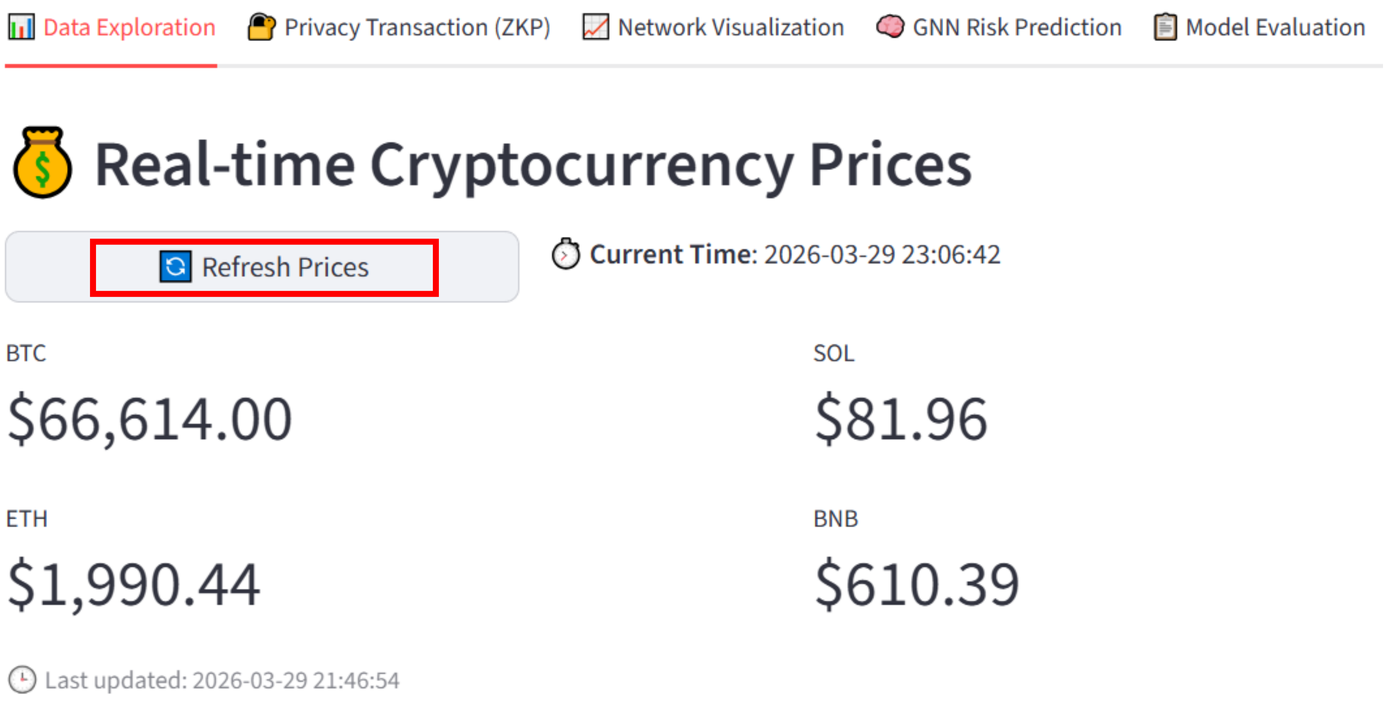
- Read from a local file (CSV or Parquet); if the file is missing, simulated data is generated automatically.
- Convert the amount field to ETH (divided by 10181018) and filter out invalid values.
- Randomly sample 500 transactions and construct an address graph (where addresses are nodes and transactions are edges).

Elliptic Data

- The feature file has no header and contains 167 columns (column 1: txId, column 2: time_step, and the remaining 166 columns: anonymous features); column names must be added manually.
- Label mapping: '1' → illegal, '2' → legal, 'unknown' → unknown (excluded during training).
- Retain only labelled nodes (valid + invalid), and split the training and test sets by time step (time steps 1–39 for training, 40–49 for testing), ensuring temporal consistency.

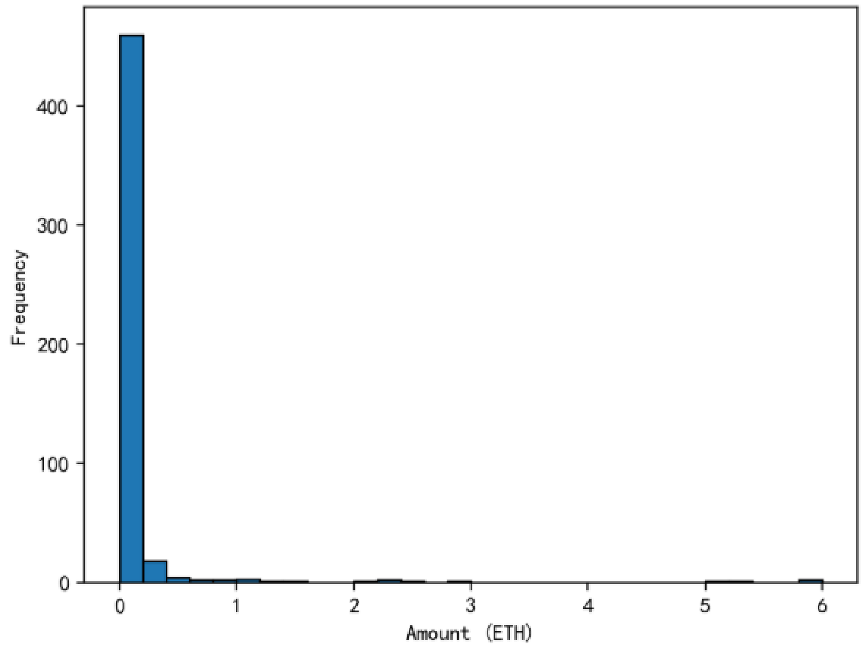
Data sources and pre-processing

Module	Data source	Features	Loading method
Real-time price dashboard	CoinGecko Free Public API	Real-time cryptocurrency prices (BTC, ETH, SOL, BNB, etc.), quoted in US dollars, returning data in JSON format. No API key required; supports manual refresh.	Call via an HTTP GET request https://api.coingecko.com/api/v3/simple/price?ids=coin_ids}&vs_currencies=usd Use the Python requests library to retrieve the data and display it dynamically in Streamlit.



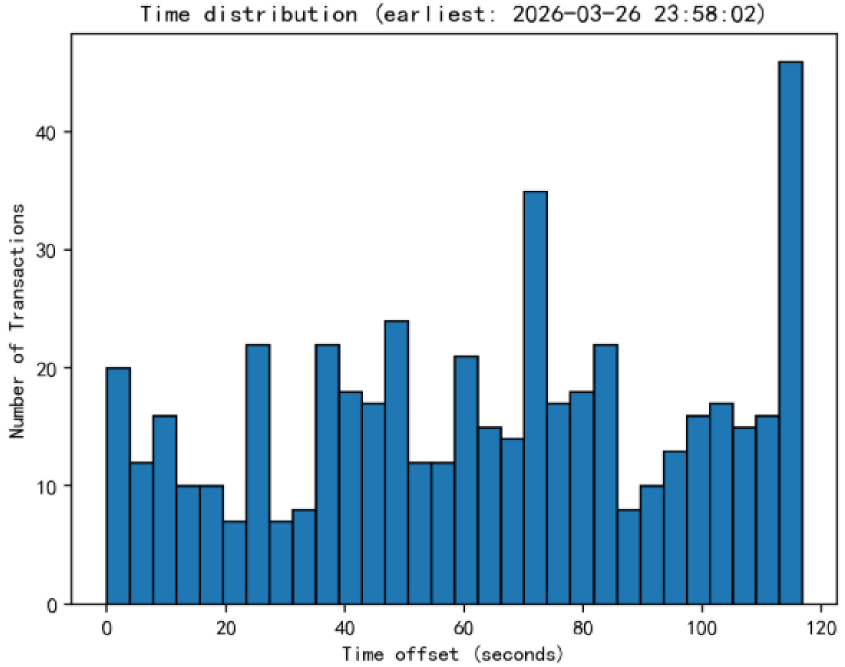
Exploratory Analysis – Flashbots Data

💰 Real Transaction Amount Distribution



Transaction Amount Distribution:
Pronounced Long-Tail Characteristics

⏰ Real Transaction Time Distribution



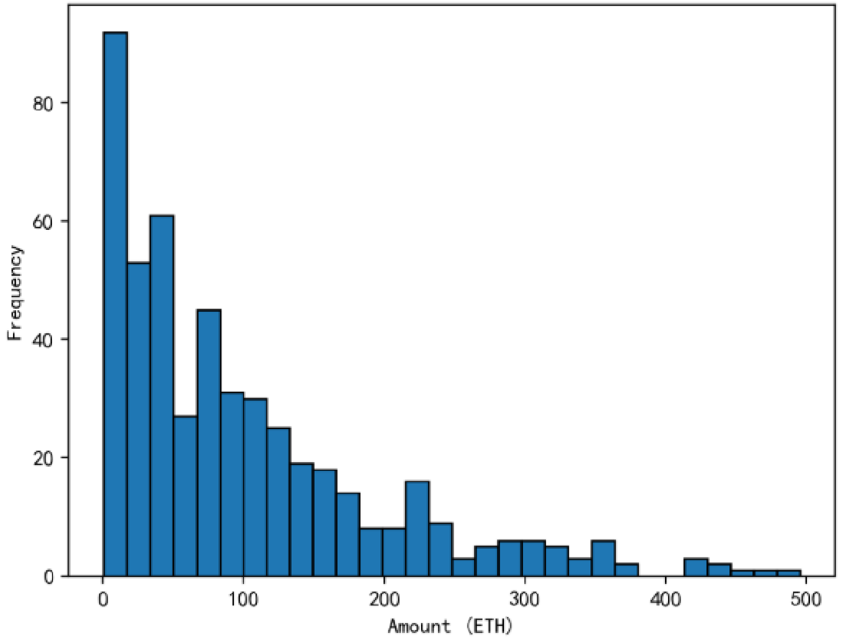
Trading Time Distribution:
Distinct Active Periods

These data features provide critical baseline references for subsequent anomaly detection models and user behavior prediction.

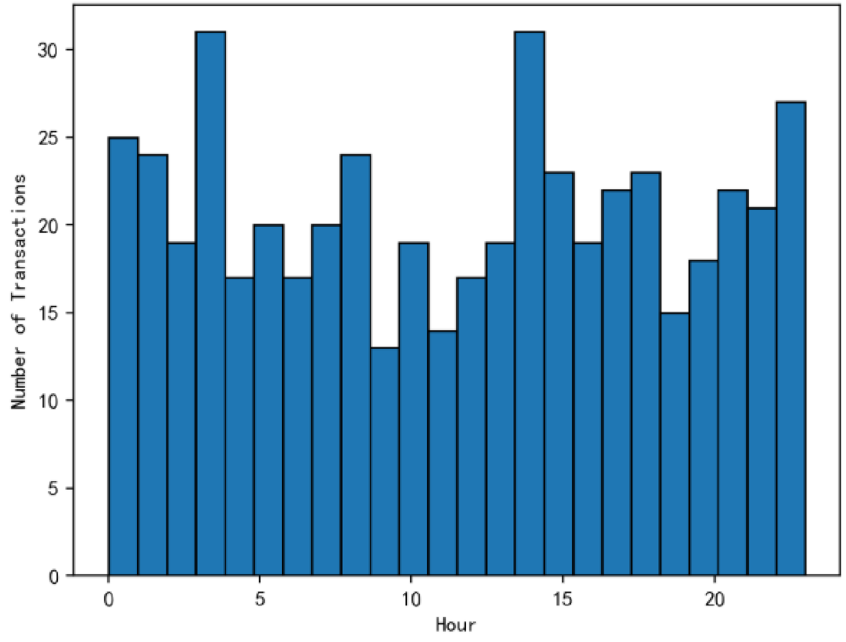
Exploratory Analysis – Flashbots Data

厦門大學 XIAMEN UNIVERSITY

💰 Simulated Transaction Amount Distribution

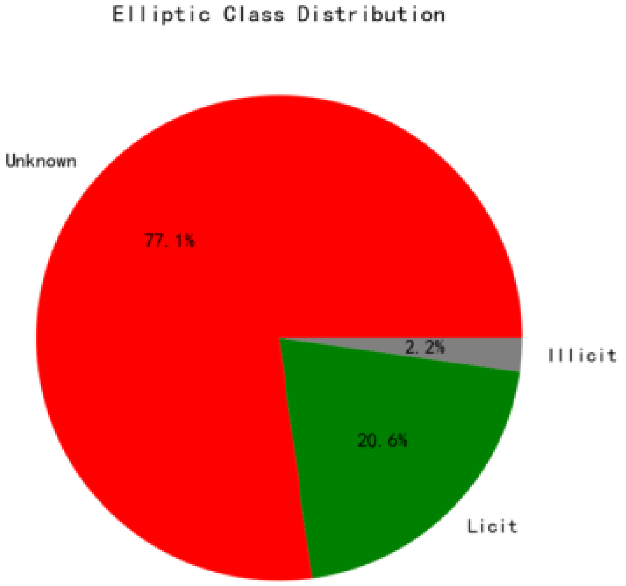


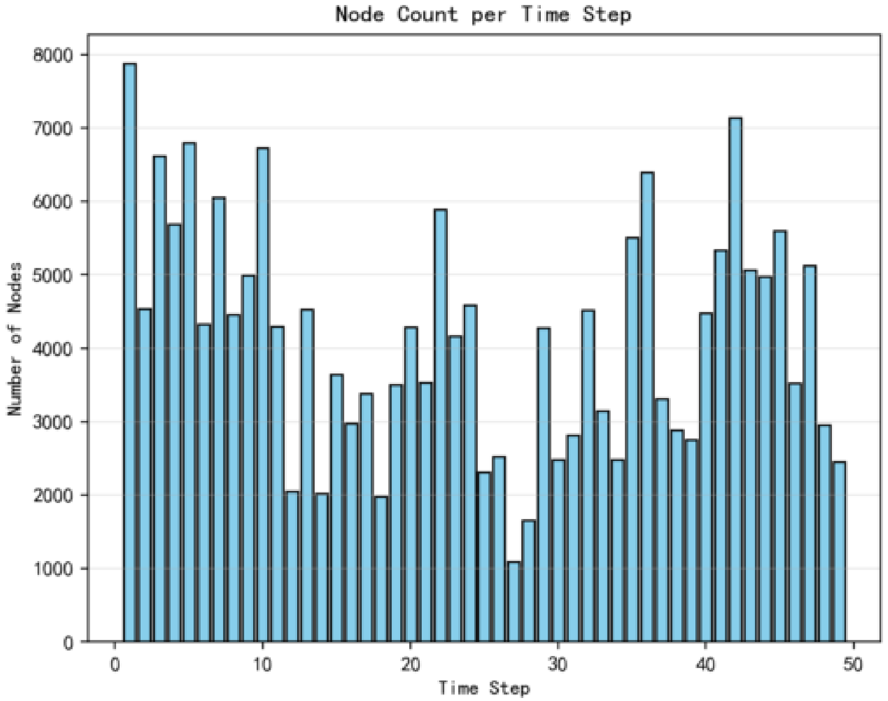
⏰ Simulated Transaction Time Distribution (Hour)

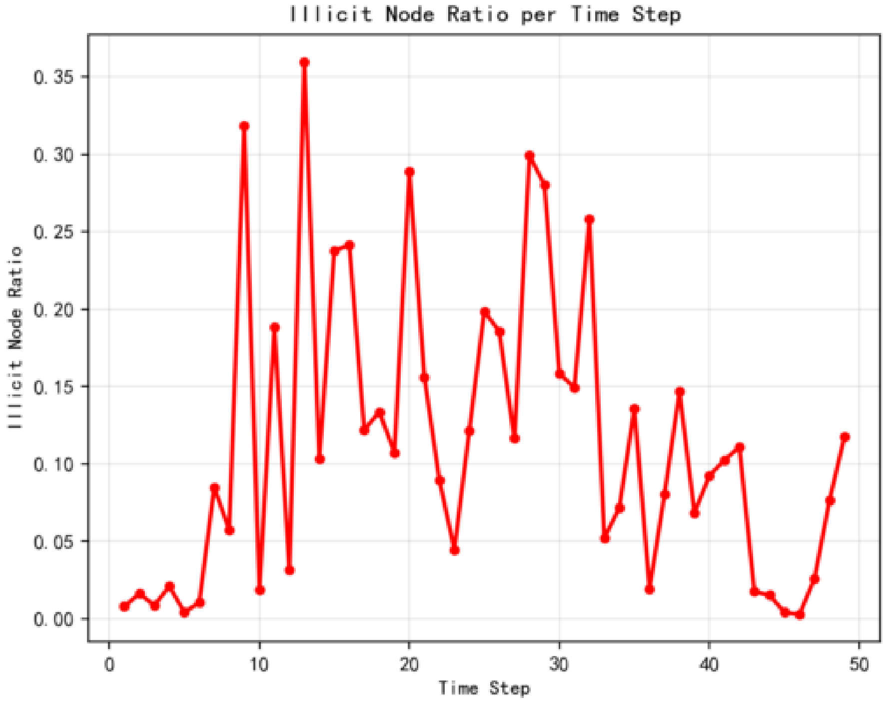


Simulated data and real data share similarities in their statistical characteristics.

Exploratory Analysis – Elliptic Data







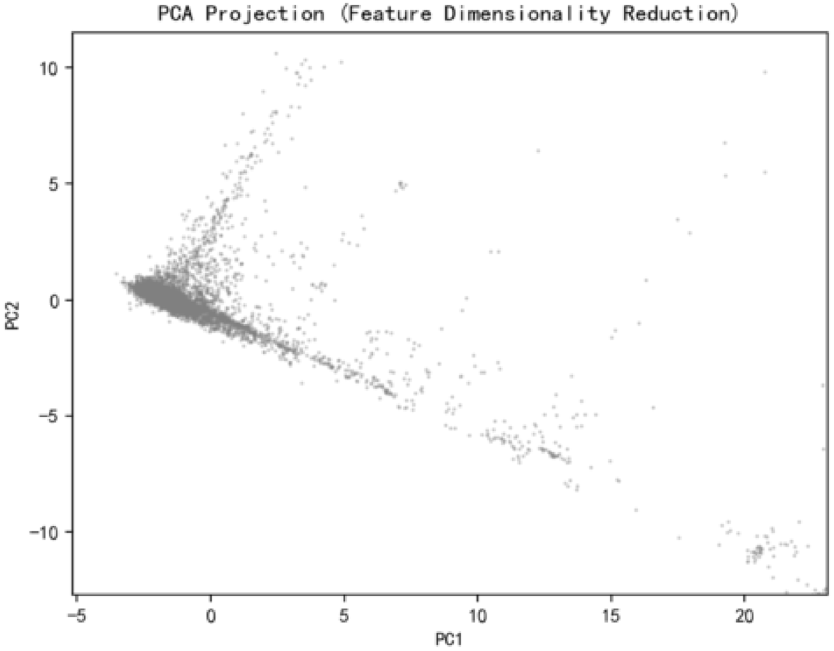
The vast majority of nodes in the dataset are unknown, with a very small proportion of legitimate and illegitimate nodes. There is a serious class imbalance problem, which is the main challenge for model training.

The number of nodes shows a dynamic trend over time, reflecting the expansion and activity fluctuations of the blockchain trading network over time.

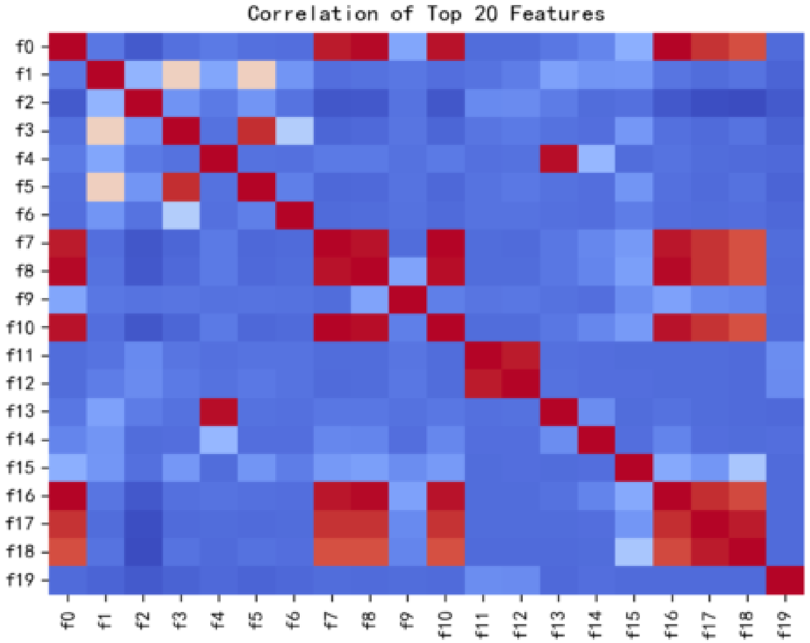
The proportion of illegal nodes exhibits significant fluctuations at different time steps, and identifying peak periods is crucial for understanding the explosive patterns of illegal activities.

Hence, the imbalance and temporal dynamics of the dataset are the core issues that must be addressed when constructing GNN fraud detection models.

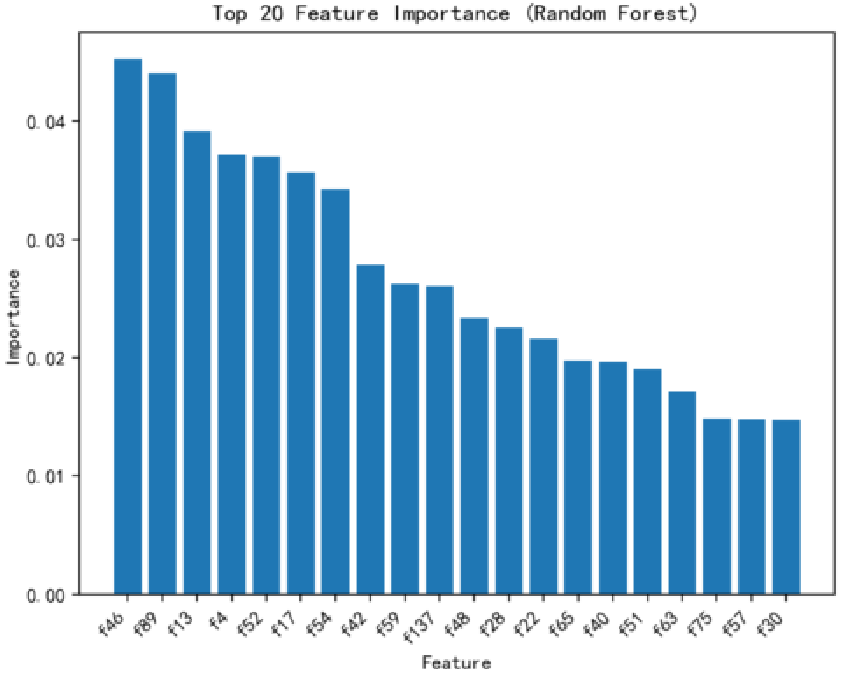
Exploratory Analysis – Elliptic Data



PCA "projects" the 166 dimensional data to the two most important directions. The point cloud tends to gather in the center and disperse around, indicating that the features have a certain structure. On this basis, graph neural network (GNN) is used to capture more complex nonlinear relationships.



This thermal map explains the relationship between features, in which blue indicates negative correlation, red indicates positive correlation, and the deeper the color, the stronger the relationship.



This figure shows another feature of the Elliptic Data, which outputs the relative importance of the first 20 features through the double random mechanism (bootstrap sampling + feature subset selection).

厦門大學
XIAMEN UNIVERSITY

03

Core Methods & System Architecture

Zero knowledge proof module

Commitment scheme:

We adopt Pedersen Commitment to meet homomorphism:

$$\mathrm{Com}(x_1,r_1)\cdot\mathrm{Com}(x_2,r_2)=\mathrm{Com}(x_1+r_1,x_2+r_2)$$

Non-interactivity:

We convert the interactive proof to a non-interactive one via the Fiat-Shamir transform, where the challenge $c = \mathrm{SHA256}(t||y)$.

Protocol:

We use Σ-protocol containing three steps:

1.Commitment: The prover randomly generates a random number r, calculates the commitment $t = g^r \bmod p$, and sends it to the verifier.

2. Challenge: The verifier (or via the Fiat-Shamir transform, simulated by a hash function) generates a random challenge c.

3. Response: The prover computes the response

$$s = r + c\cdot x \bmod (p-1).$$

Modulus Selection:

Provides two options: $p = 97$ (demo-level) and $p = 2^{256}-2^{32}-977$ (industrial-grade, secp256k1 curve). The demo-level modulus facilitates numerical display, while the industrial-grade one ensures practical security.

Neural Network Module

◆ **Graph Neural Network(GNN) Model:**

It realizes the aggregation of node features through "graph convolution operation": the weighted sum of the neighborhood features of nodes is fused with their own features, which is essentially the convolution (spectral domain/spatial domain) of the graph's adjacency matrix.

In our project, it is used to analyze the fraud risk of nodes (addresses) in the blockchain transaction graph.

For imbalanced categories (less fraud samples and more legitimate samples in blockchain), category weight is added to cross entropy loss.

- **Category weight:** Give higher weight to the "fraud category " with less samples to avoid the model bias to the majority category (legal category).
- **The loss function:**

$$\text{CrossEntropyLoss} = -\Sigma\,(\text{y\_true} * \log(\text{y\_pred}))$$

Here, y_true is the true label of the node (0=legal, 1=fraud), and y_pred is the probability of model output.

Neural Network Module

- **Graph Convolutional Network(GCN) Model:**

  It is the core subclass of GNN, and realizes the aggregation of node features through "graph convolution operation": the weighted sum of the neighborhood features of nodes is fused with their own features, which is essentially the convolution (spectral domain/spatial domain) of the graph's adjacency matrix.

In our project, based on the risk detection model of GCN, it extracts the node characteristics of the transaction graph through multi-layer gcnconv.

- **Graph Attention Network(GAT) Model:**

  It is another important subclass of GNN. Based on GCN, attention mechanism is introduced: different attention weights are assigned to different neighbor nodes (rather than fixed neighbor weights of GCN), which can adaptively capture the characteristics of important neighbors and improve the modeling ability of key nodes.

Technically, gat is improved based on GCN, which solves the problem of "treating all neighbors equally" in GCN, and highlights important neighbors through attention weight.

Neural Network Module

Graph structure: In Elliptic data, nodes represent transactions and edges represent the flow of Bitcoin, constructing a directed graph.

GCN: Graph Convolutional Network learns node representations through neighbor aggregation.

GAT: Graph Attention Network, introducing multi head attention mechanism to weight neighbor contributions.

Loss function: Weighted cross entropy loss, where the weight is inversely proportional to the category frequency, alleviates category imbalance (illegal nodes are only 2%).

Training strategy: Adam optimizer, initial learning rate of 0.001, decay of 0.5 every 30 rounds, patient early stop for 20 rounds.

Evaluation indicators: Accuracy, precision, recall, F1 score ROC AUC.

Overall system architecture

厦門大學
XIAMEN UNIVERSITY

Application Layer (Streamlit)	Logic Layer (Python)	Data Layer (Pandas/PyG)
• **Data Exploration:** Flashbots/Elliptic Statistics and Visualization • **Privacy transactions**: ZKP proof generation and verification, homomorphic demonstration • **GNN training:** model configuration, training curve, evaluation metrics, prediction results • **Network visualization:** dynamic coloring of real/simulated trading networks, highlighting of risk nodes	• ZKP proof engine • GNN risk predictor • ∑ Protocol • GCN/GAT • Pedersen Commitment • Weighted Cross Entropy Loss • Fiat Shamir • Early Stop&Learning Rate Scheduling	• Flashbots Local Files/Simulated Data • Elliptic local files+feature standardization • Graph Structure Construction (NetworkX → PyG Data)

04

Experimental Setup & Process

Experiment environment

Hardware configuration:

Cpu: Intel i7 processor

Memory: 16GB DDR4

GPU: NVIDIA RTX 3060

Software:

Python 3.9 , PyTorch 2.0 , Streamlit 1.2, PyTorch Geometric 2.3 , Pandas, NumPy, Matplotlib, Seaborn, scikit-learn

Experiment process

Data preparation → **System startup** → **Interaction and observation**

Prepare real transaction data and elliptic data sets and place them in the specified record structure.

At the terminal, execute the command 'streamlit run zkp_gnn_viz.py' to start visualization.

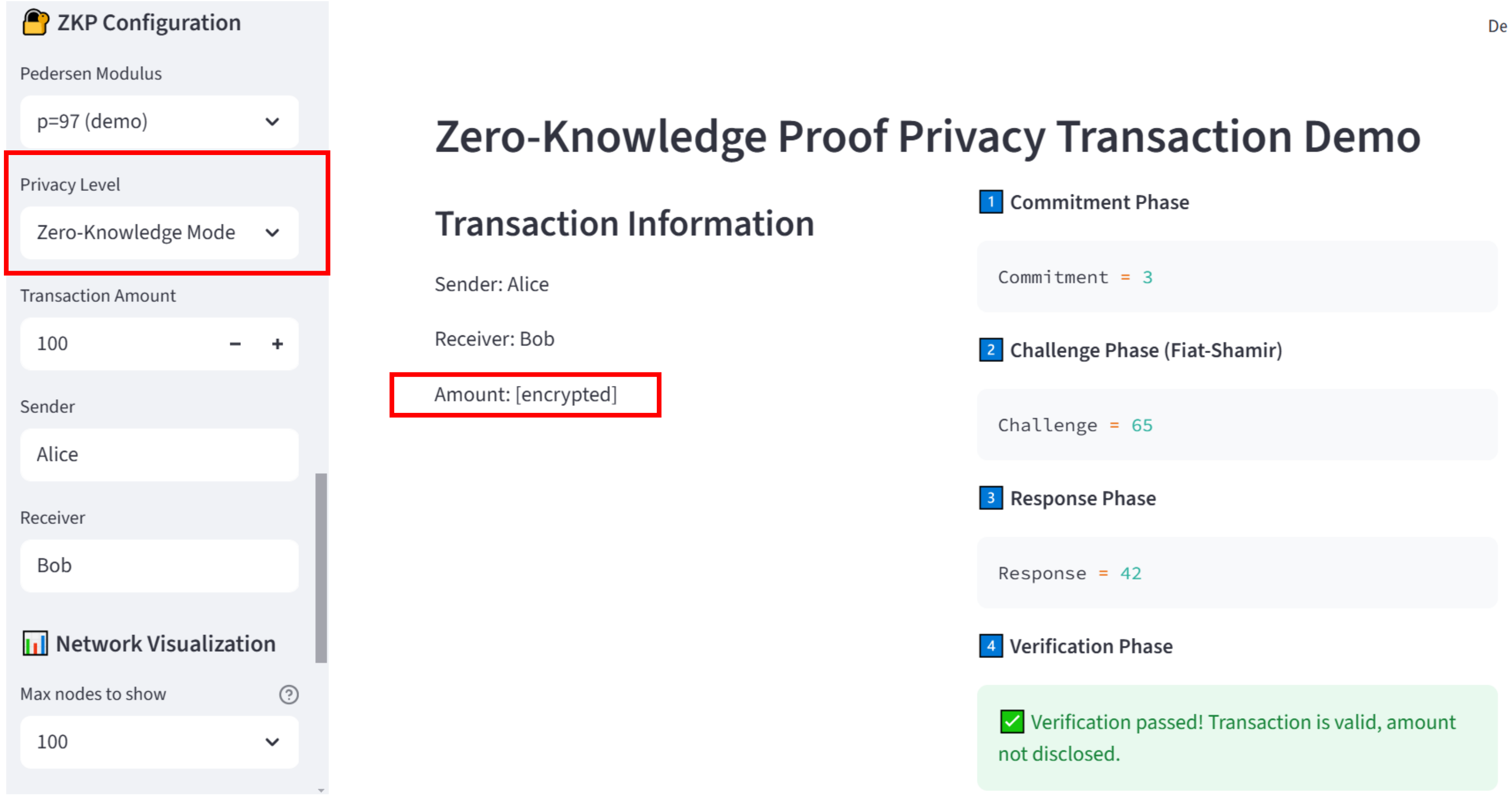
In the browser, data exploration, zkp evolution and GNN model evaluation are carried out, and the results are recorded.

05

Result Display & Analysis

Demonstration of zero knowledge proof

After the user enters the sender "Alice", receiver "Bob" and the amount of 100, and selects "zero knowledge mode", the system automatically generates a certificate and verifies it. (Take p=97 as an example.)



Demonstration of zero knowledge proof

◆ ZKP—Commitment & Challenge

Item	Content
Scenario	Alice → Bob, Amount = 100, Privacy Mode = ON
System Parameters	p=97 (prime modulus), g=5 (generator), h=7 (second generator), order=p−1=96 (order of multiplicative group), r (nonce)
Secret	x=100 mod 96=4 (hidden secret value)

Demonstration of zero knowledge proof

◆ ZKP—Commitment & Challenge

$$C = g^{x} \cdot h^{r} \,(mod\ \ p)$$

Commitment Phase: The commitment C is computed by multiplying g^x and h^r, then reducing modulo p, where x is the secret value and r is a random nonce.

$$c = Hash(C \,||\, PublicKey)\,(mod\ order)$$

Challenge Phase (Fiat-Shamir): The challenge c is obtained by hashing the concatenation of the commitment C and the public key g^x, then reducing modulo the group order.

Demonstration of zero knowledge proof

◆ ZKP—Response, Verification & Homomorphism

$$s = r + c \cdot x \,(mod\ order)$$

Response Phase: The response s is the sum of the random nonce r and the product of the challenge c and the secret x, reduced modulo the group order.

$$g^s \stackrel{?}{=} C \cdot (g^x)^c \,(\mathrm{mod}\ p)$$

Verification Phase:The verifier checks whether g^s modulo p equals the product of the commitment C and the public key g^s raised to the power c, modulo p.

Demonstration of zero knowledge proof

Demonstration of homomorphic properties:

Commitment(A) = 30

Commitment(B) = 20

And we can see the homomorphism verification is ‘True’.

Pedersen Commitment Homomorphism

Amount A

30 − +

Amount B

20 − +

Commitment(A): 28

Commitment(B): 81

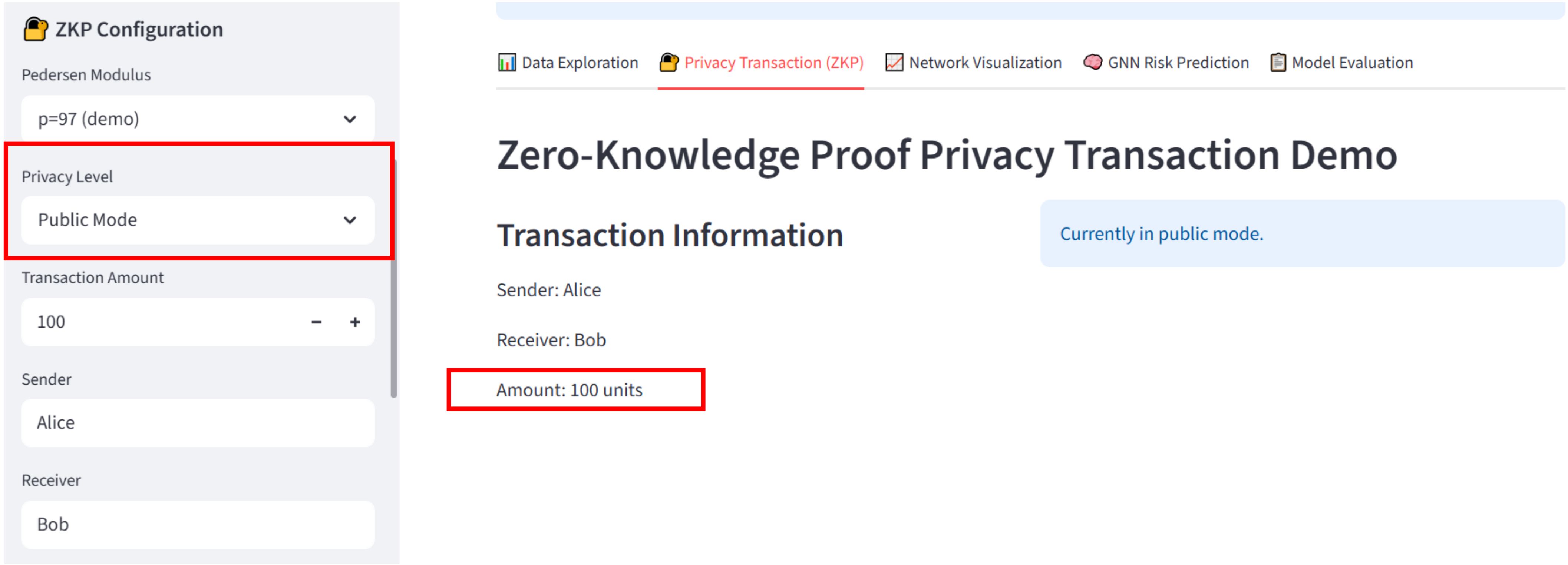
Commitment(A)+Commitment(B): 37

Direct Commitment(A+B): 37

Homomorphism verification: True

Demonstration of zero knowledge proof

Once we change the privacy setting to ‘public mod’, the system will display the amount directly. (Take p=97 as an example.)



GNN model prediction results

After the training, the system outputs the following evaluation indicators (taking the comparison between GCN and gat as an example):

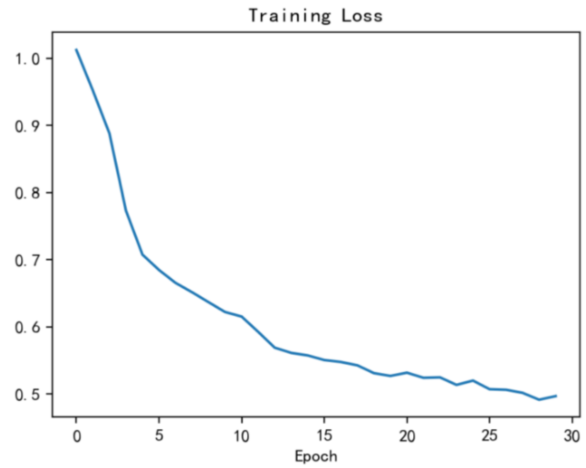
Model	Accuracy	Precision	Recall	F1 Score	AUC
GCN	75.65%	27.21%	89.22%	41.70%	0.897
GAT	81.04%	33.24%	93.51%	49.05%	0.948

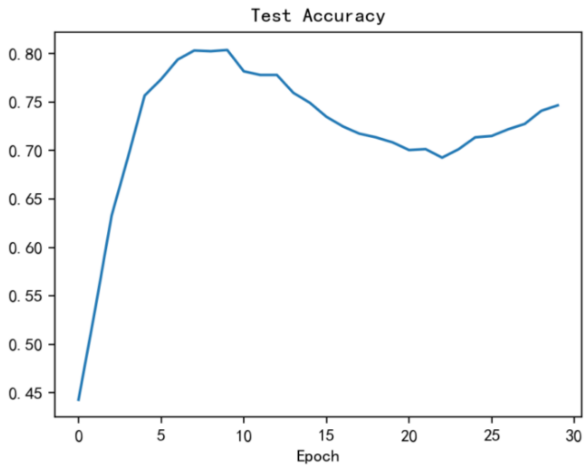
It should be noted that the reproducibility of results depends on the **random seed** set in the programme.

GNN model prediction results

GCN Results

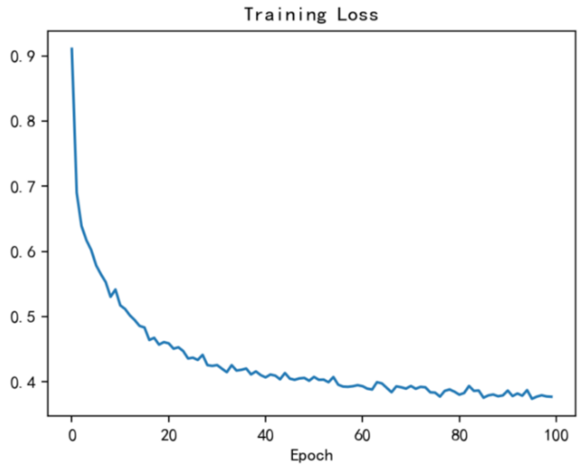
Accuracy	Precision	Recall	F1	AUC
75.65%	27.21%	89.22%	41.70%	0.897

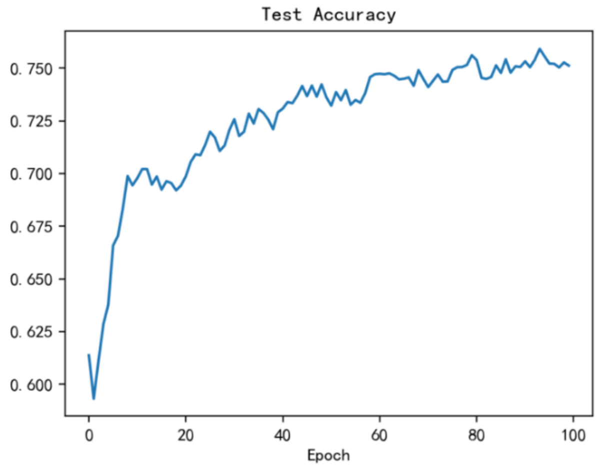




GAT Results

Accuracy	Precision	Recall	F1	AUC
81.04%	33.24%	93.51%	49.05%	0.948

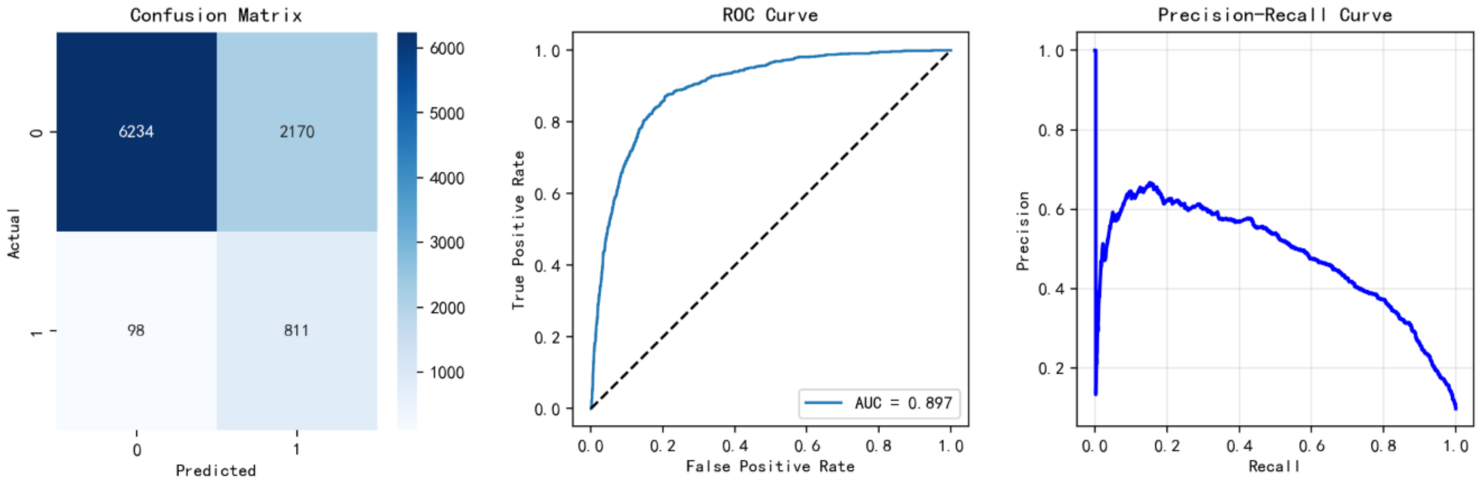




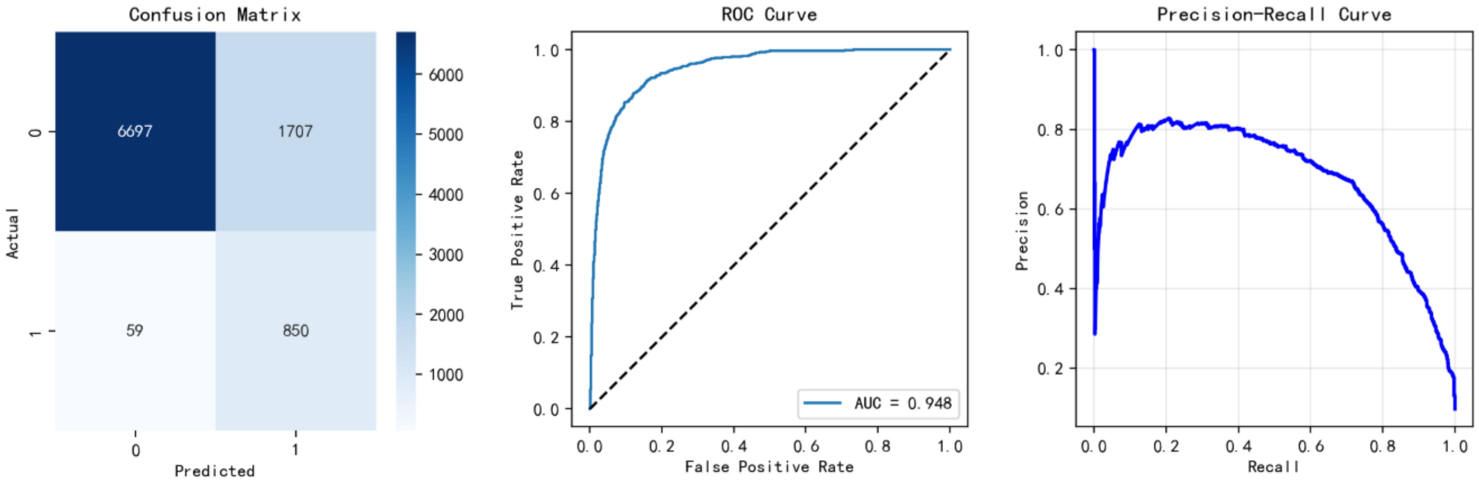
GCN Results	GAT Results
GCN converges after approximately 5 iterations.The training loss decreases rapidly, but the test accuracy peaks early and then declines, indicating potential overfitting or unstable generalization after a certain training stage.	**GAT** converges more quickly due to its attention mechanism.The training loss decreases steadily over 100 epochs, and the test accuracy shows consistent improvement, suggesting stable convergence without severe overfitting.

GNN model prediction results

GCN Evaluation



GAT Evaluation



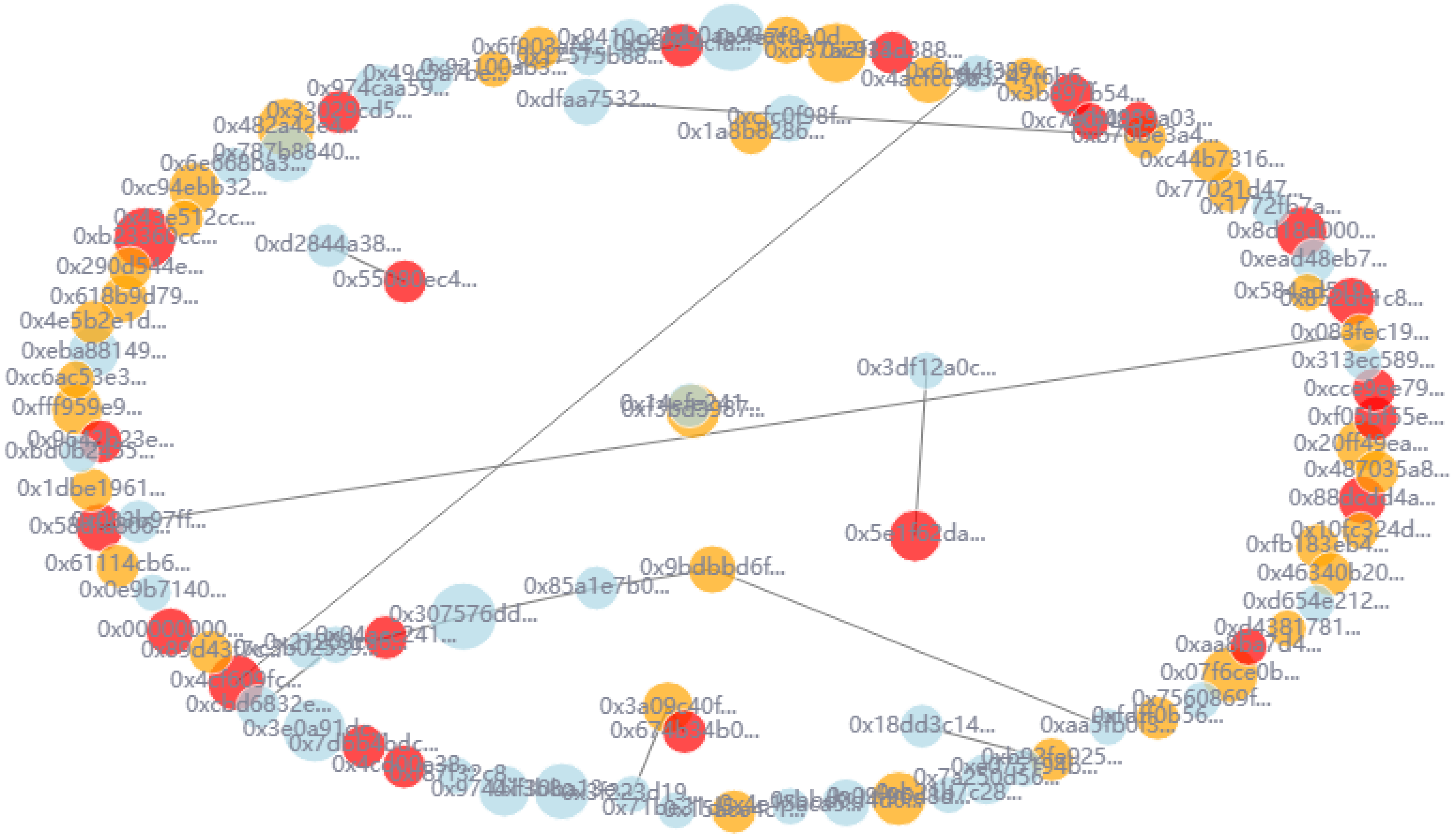
GCN misclassified 10.78% of illegal nodes and GAT reduced this to 6.49% via attention weights,with a higher AUC (0.948 vs 0.897).

Network visualization

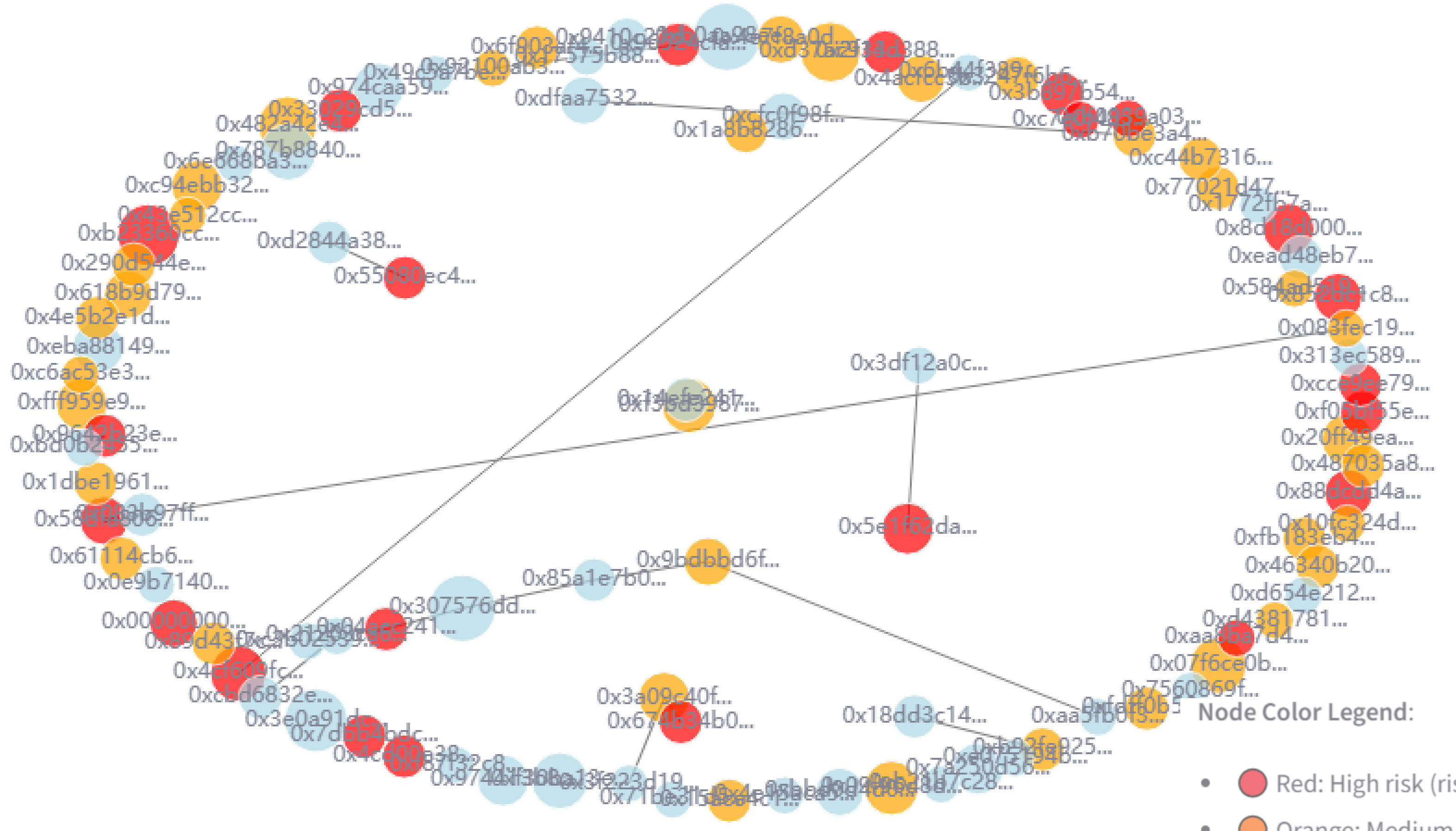
Here is the **real transaction network**.

It shows the transaction relationship between addresses, and the node color is random.

The example image shows 100 nodes; the system offers a choice of 50, 100, 200 or all nodes.



Network visualization



Node Color Legend:

- Red: High risk (risk score > 0.7)
- Orange: Medium risk (0.3 < risk score ≤ 0.7)
- Blue: Low risk (risk score ≤ 0.3)

Node Size: Larger nodes represent higher degree (more connections).

Edge Width: Thicker edges indicate larger total transaction amount between the same pair of addresses.

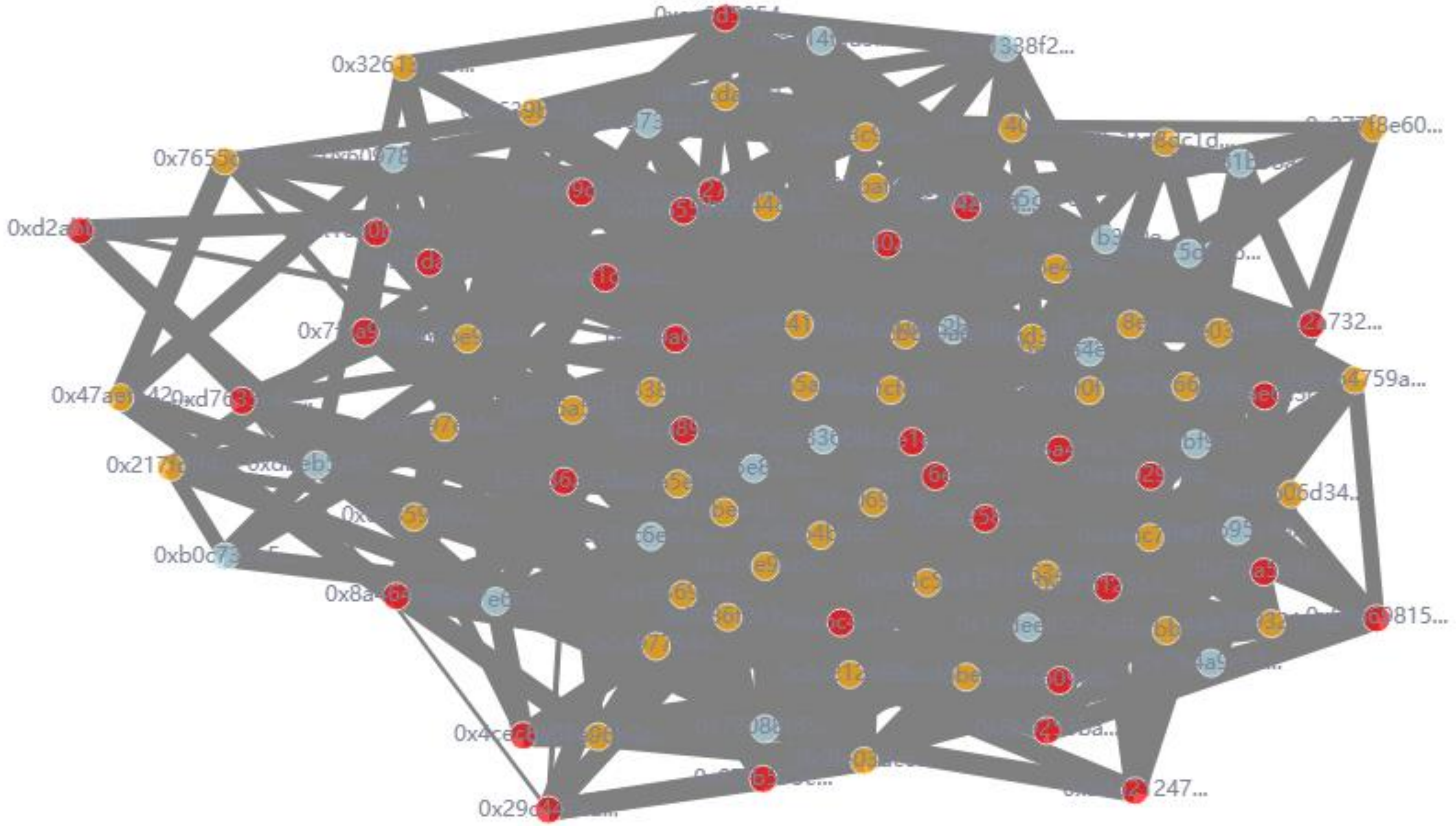
Note: Node colors are randomly generated risk scores for demonstration only.

Network visualization

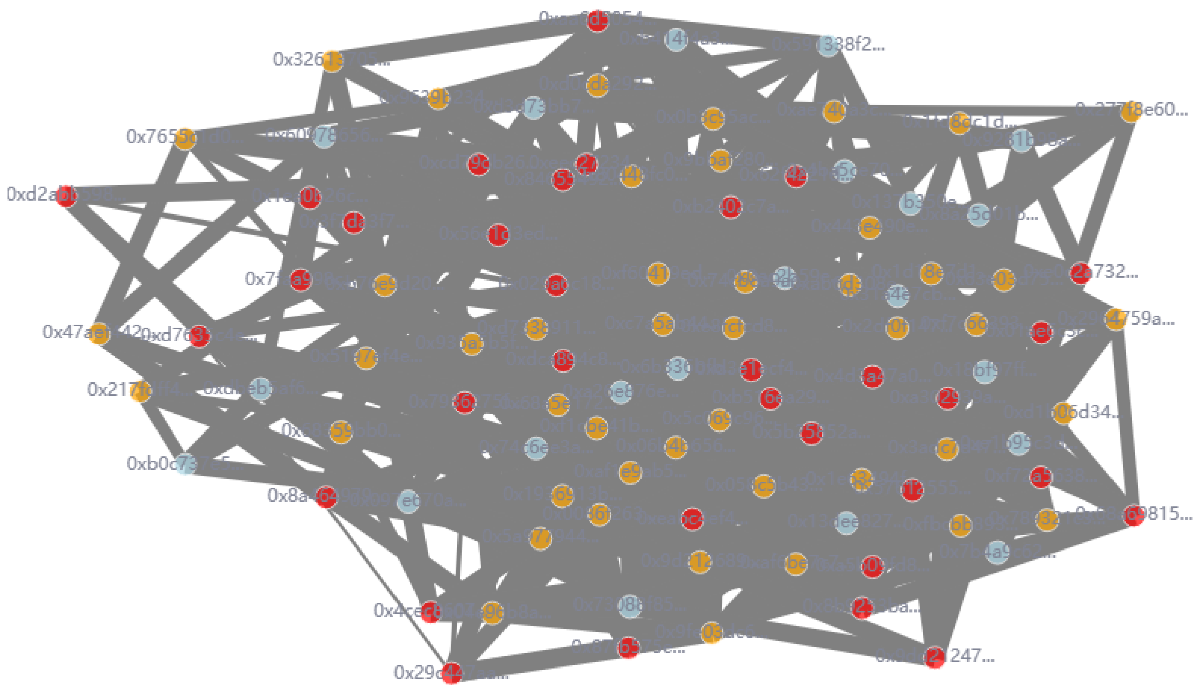
Here is the **simulated transaction network**.

Its structure is similar to the real network, but the colors of the nodes are random.

The example image shows 100 nodes; the system offers a choice of 50, 100, 200 or all nodes.



Network visualization



Node Color Legend:

- Red: High risk (risk score > 0.7)
- Orange: Medium risk (0.3 < risk score ≤ 0.7)
- Blue: Low risk (risk score ≤ 0.3)

Node Size: Larger nodes represent higher degree (more connections).

Edge Width: Thicker edges indicate larger total transaction amount between the same pair of addresses.

Note: This network is based on simulated transaction data. Node colors are randomly generated risk scores for visualization demonstration only.

06

Summary & Future Work

Summary

ZKP Module

- **Zero-knowledge proof demonstration:**
  Through a simplified implementation of the Σ protocol, the core logic of the 'promise-challenge-response' scheme has been successfully demonstrated; the Fiat-Shamir transformation converts interactive proofs into non-interactive ones, making them suitable for blockchain environments.
- **Pedersen's homomorphic verification:**
  The verification of homomorphic properties provides an intuitive demonstration of the additive homomorphism of the Pedersen commitment, offering a clear illustration for understanding the verification of aggregated amounts in private transactions.
- **Scalability of cryptographic applications:**
  The modulus switching function demonstrates the scalability of cryptography from educational demonstrations to industrial-grade applications.

Summary

厦門大學
XIAMEN UNIVERSITY

GNN Module

- **Loss functions to mitigate class imbalance:**
  The Elliptic dataset suffers from severe class imbalance (with illegal nodes accounting for only 2% of the data); the use of weighted cross-entropy loss can effectively mitigate the model’ s tendency to favour the majority class.
- **GAT models outperform GCNs:**
  The GAT model outperforms the GCN on certain metrics (such as recall and F1 score), indicating that the attention mechanism is more effective at identifying key neighbour features, which has a positive impact on minority class detection.
- **Anonymous features can still be distinguished:**
  Although feature anonymisation limits interpretability analysis, PCA and random forest feature importance analyses still reveal the structural properties of the features, indicating that anonymised features retain their discriminatory power.
- **Temporal scaling enhances performance potential:**
  Time-step information is not directly used as GNN input (it is used solely for splitting the data into training and testing sets); in future, a temporal graph network could be introduced to further improve performance.

Summary

厦門大學 XIAMEN UNIVERSITY

System Integration and Visualisation

- **Intuitive and easy to use:**
  The interactive interface built with Streamlit seamlessly integrates complex technical modules, allowing users to explore blockchain data, run models and observe the results without the need for programming.
- **Dual data for guaranteed availability:**
  By displaying real data and simulated data in parallel, the system ensures availability during periods of network instability whilst maintaining the reliability of the real data.
- **Multi-dimensional visualisation support:**
  Multi-dimensional visualisation (value distribution, temporal distribution, network topology, feature space, model evaluation) comprehensively addresses data exploration and model validation requirements.

Summary

厦門大學
XIAMEN UNIVERSITY

Conclusion

This project has successfully developed a blockchain-based privacy protection and risk prediction visualisation system that integrates zero-knowledge proofs with graph neural networks.

The main achievements include:

◆ A non-interactive zero-knowledge proof module based on the Σ protocol, capable of demonstrating the verification process for private transactions and supporting the verification of homomorphic properties.

◆ A GNN-based risk prediction module using the Elliptic dataset compared the performance of GCN and GAT, thereby validating the effectiveness of graph neural networks in detecting fraudulent transactions.

◆ A comprehensive Streamlit visual interface that integrates a range of features, including real-data exploration, simulated data comparison, network topology visualisation, and model training and evaluation.

Limitations and Future Work

厦門大學 XIAMEN UNIVERSITY

Limitations

◆ **Data heterogeneity:** The ZKP module uses Ethereum transaction data (Flashbots), whilst the GNN module uses Bitcoin transaction data (Elliptic). Although these are not the same blockchain, the modules operate independently and do not affect the purpose of the demonstration.

◆ **Risk score decoupling:** Risk scores in network visualisation are currently generated randomly and are not linked to GNN prediction results; in future, an address-transaction mapping could be established to feed the model' s predicted risk back into the network graph.

◆ **Real-time:** Flashbots' real-time data must be downloaded manually by users; automatic online retrieval is not yet available (due to restrictions on access to data sources), but a workaround using simulated data is provided.

Limitations and Future Work

Future Work

- **Integration of risk scores with network visualisation:** mapping the risk scores predicted by the GNN back to address nodes to enable dynamic colouring of risk heatmaps.
- **Incorporating temporal information:** By utilising the temporal step information in the Elliptic dataset, we construct a temporal graph neural network (such as T-GCN) to capture dynamic evolutionary patterns.
- **Expanding data sources:** Unify ZKP and GNN data sources; for example, train the GNN using transaction data from the same blockchain (Ethereum) to enhance the relevance of the modules.
- **Improving interpretability:** Use GNNExplainer or an ensemble of feature importance analyses to explain the basis for the model' s predictions.
- **Deployed as an online service:** The system is deployed on a cloud platform and supports real-time transaction risk alerts.

THE END

Thanks for watching!

References :

[1] Goldwasser, S., Micali, S., & Rackoff, C. (2019). The knowledge complexity of interactive proof-systems. In Providing sound foundations for cryptography: On the work of shafi goldwasser and silvio micali (pp. 203-225).

[2] Pedersen, T. P. (1991, August). Non-interactive and information-theoretic secure verifiable secret sharing. In Annual international cryptology conference (pp. 129-140). Berlin, Heidelberg: Springer Berlin Heidelberg.

[3] Fiat, A., & Shamir, A. (1986, August). How to prove yourself: Practical solutions to identification and signature problems. In Conference on the theory and application of cryptographic techniques (pp. 186-194). Berlin, Heidelberg: Springer Berlin Heidelberg.

[4] Kipf, T. N., & Welling, M. (2016). Semi-supervised classification with graph convolutional networks. arXiv preprint arXiv:1609.02907.

[5] Veličković, P., Cucurull, G., Casanova, A., Romero, A., Lio, P., & Bengio, Y. (2017). Graph attention networks. arXiv preprint arXiv:1710.10903.

[6] Weber, M., Domeniconi, G., Chen, J., Weidele, D. K. I., Bellei, C., Robinson, T., & Leiserson, C. E. (2019). Anti-money laundering in bitcoin: Experimenting with graph convolutional networks for financial forensics. arXiv preprint arXiv:1908.02591.

[7] Bünz, B., Bootle, J., Boneh, D., Poelstra, A., Wuille, P., & Maxwell, G. (2018, May). Bulletproofs: Short proofs for confidential transactions and more. In 2018 IEEE symposium on security and privacy (SP) (pp. 315-334). IEEE.

[8] Nicholls, J., Kuppa, A., & Le-Khac, N. A. (2023, December). Fraudlens: Graph structural learning for bitcoin illicit activity identification. In Proceedings of the 39th Annual Computer Security Applications Conference (pp. 324-336).

[9] Jeyakumar, S. T., Eugene Yugarajah, A. C., Hóu, Z., & Muthukkumarasamy, V. (2023, November). Detecting malicious blockchain transactions using graph neural networks. In International symposium on distributed ledger technology (pp. 55-71). Singapore: Springer Nature Singapore.